IBM SPSS Modeler 18.0

Telecom Customer Churn Prediction — Practical Tasks

Updated print layout with larger screenshots and expanded step-by-step descriptions.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Step 1 of 5



Opening the working-directory dialog. The Set Directory dialog is opened from IBM SPSS Modeler. This dialog lets the user choose the folder where the project files and related outputs will be stored.

Illustrative mockup; not a genuine capture from a running IBM SPSS Modeler session.

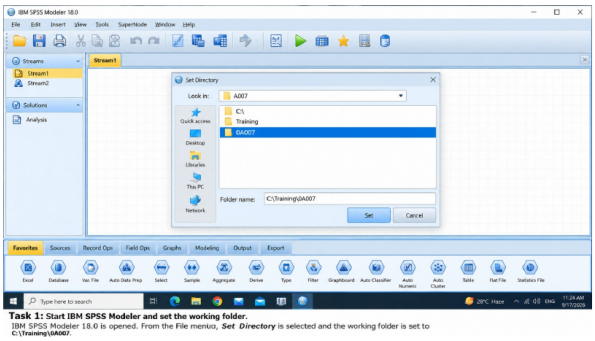
Objective: Establish the working folder before building the data-mining stream. In IBM SPSS Modeler, the working directory is the default location used to find and save project files when relative paths are used.

Procedure: Open the directory-setting option from the application menu. The directory dialog is where the folder for this practical is selected. Use the folder specified by your instructor or the project instructions, and ensure that the folder exists before proceeding.

Why it matters: Setting the correct folder at the beginning helps keep the source workbook, saved stream, and exported prediction file together. It also reduces errors caused by loading files from the wrong location.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Step 2 of 5

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 2 of 5



Task 1: Start IBM SPSS Modeler and set the working folder.
IBM SPSS Modeler 18.0 is opened. From the File menu, ***Set Directory*** is selected and the working folder is set to `C:\Training\0A007`.

Choosing the project folder. The folder path shown is C:\\Training\\0A007. Selecting this directory keeps the project data and exported files organized in one working location.

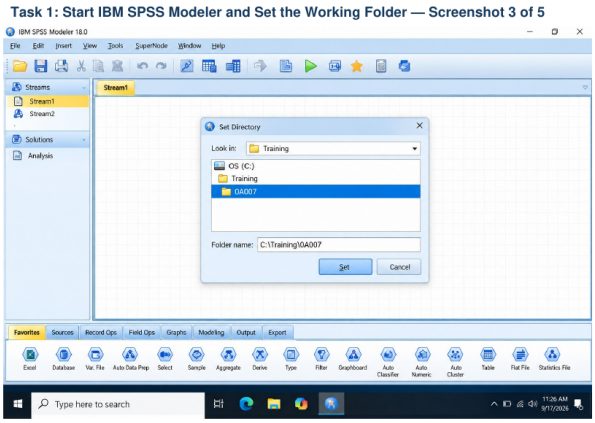
Illustrative mockup; not a genuine capture from a running IBM SPSS Modeler session.

Objective: Choose the folder that will hold the practical's input and output files.

Procedure: Browse to the project folder under the Training directory and select it. The example path shown in this sequence is `C:\Training\0A007`; check the spelling of the folder name against your own course instructions because a path must match the folder on your computer exactly.

Check: Confirm that the selected folder is the intended project location and that you have permission to read and write files there. Keep the original Excel workbook in this folder or note its actual location so it can be selected in the source node later.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Step 3 of 5



Locating the folder in the directory browser. The directory browser displays the available folders under Training. The required project folder is selected so that it can be used as the active working directory.

Illustrative mockup; not a genuine capture from a running IBM SPSS Modeler session.

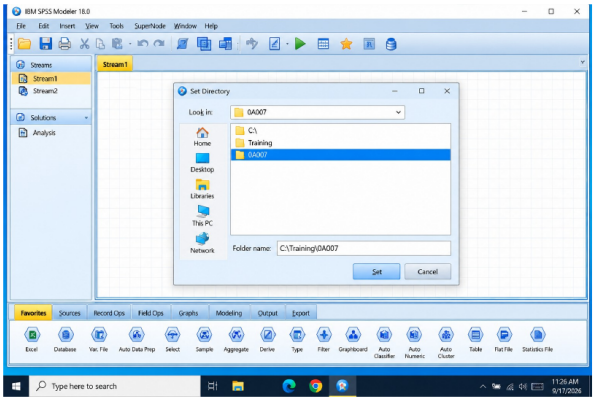
Objective: Locate the project directory in the folder browser.

Procedure: Expand the relevant parent folder, navigate through the available directories, and highlight the folder for this practical. Avoid selecting a similarly named folder or a parent folder unless the instructions specifically call for it.

Check: The selected directory should be visible in the browser before you accept the selection. This is a useful point to verify the folder name and path so that later file references are consistent throughout the workflow.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Step 4 of 5

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 4 of 5



Reviewing the selected directory. The dialog shows the selected folder before confirmation. Checking the path helps ensure that the project uses the intended working location rather than a different folder.

Illustrative mockup; not a genuine capture from a running IBM SPSS Modeler session.

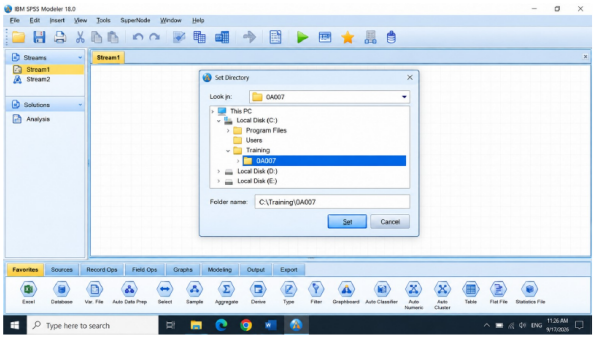
Objective: Review the directory selection before applying it.

Procedure: Inspect the displayed path and make sure it points to the intended project folder. If the path is incorrect, return to the browser and select the correct location rather than continuing with an uncertain setup.

Expected result: The working-directory dialog should show the folder you intend to use. Correct directory setup is a small but important preparation step because the stream, source data, and exported at-risk customer list need predictable file locations.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Step 5 of 5

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 5 of 5



Confirming the working-folder setup. This view shows the selected working directory in the Modeler workspace. After confirming the setting, subsequent project files can be accessed relative to this working folder.

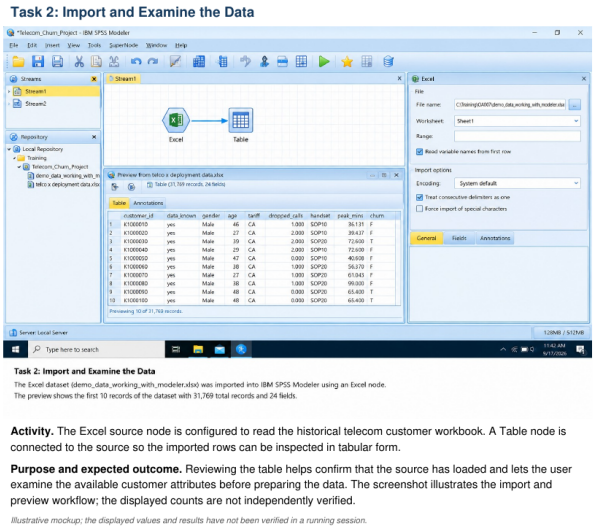
Illustrative mockup; not a genuine capture from a running IBM SPSS Modeler session.

Objective: Finish the working-folder setup and return to the Modeler workspace.

Procedure: Confirm the selected directory. After the setting is applied, continue with the data-source and modeling steps, using the selected folder as the common location for project files where appropriate.

Expected result: The workspace is ready for the next activity: importing the historical telecom customer dataset. Before moving on, verify the path once more if the source workbook cannot be found.

Task 2: Import and Examine the Telecom Customer Data



Activity. The Excel source node is configured to read the historical telecom customer workbook. A Table node is connected to the source so the imported rows can be inspected in tabular form.

Purpose and expected outcome. Reviewing the table helps confirm that the source has loaded and lets the user examine the available customer attributes before preparing the data. The screenshot illustrates the import and preview workflow; the displayed counts are not independently verified.

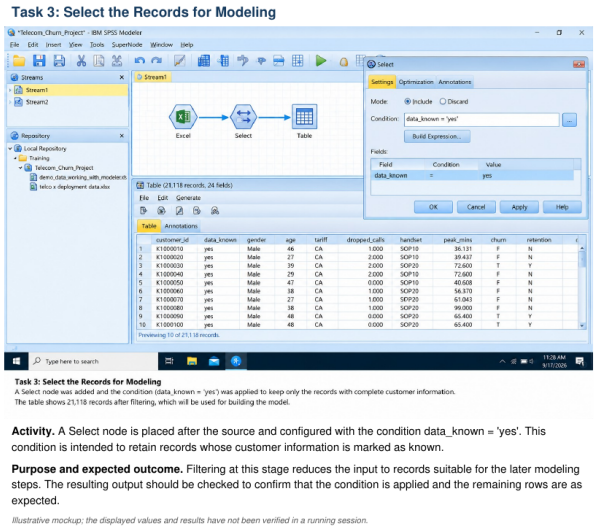
Illustrative mockup; the displayed values and results have not been verified in a running session.

Objective: Load the historical telecom customer workbook and inspect its structure before modeling.

Procedure: Add an Excel source node to the stream, browse to `demo_data_working_with_modeler.xlsx`, and configure the correct worksheet or range if prompted. Connect a Table node to the source and run it to preview the records and fields.

What to inspect: Check that the expected customer attributes and the `churn` field are present, values are aligned under the correct columns, and the data types appear reasonable. This inspection can reveal missing fields, incorrect sheet selection, or import issues before the modeling steps begin.

Task 3: Select Records with Known Customer Information



Activity. A Select node is placed after the source and configured with the condition data_known = 'yes'. This condition is intended to retain records whose customer information is marked as known.

Purpose and expected outcome. Filtering at this stage reduces the input to records suitable for the later modeling steps. The resulting output should be checked to confirm that the condition is applied and the remaining rows are as expected.

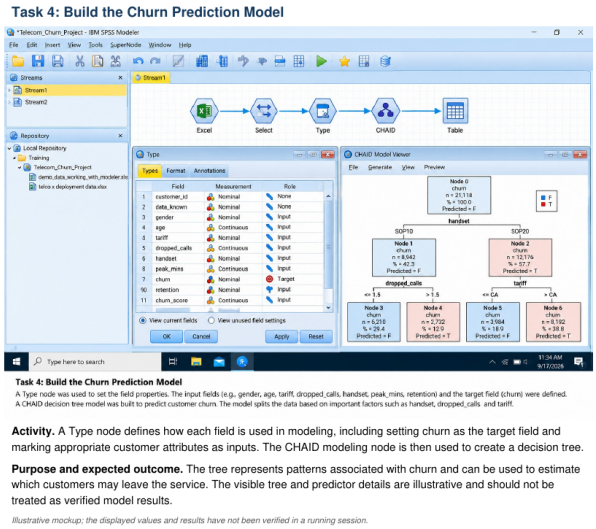
Illustrative mockup; the displayed values and results have not been verified in a running session.

Objective: Keep records whose customer information is marked as known for use in the modeling workflow.

Procedure: Place a Select node after the source node and configure its condition as `data_known = 'yes'`. Connect the Select node to a Table node or another inspection step and run the stream to review the filtered output.

Expected result: Only records matching the condition should pass through the Select node. Compare the output with the original dataset and confirm that the field name and value are spelled exactly as they appear in the data. If the row count differs from the tutorial, investigate the input file and selection condition rather than assuming the count is correct.

Task 4: Define Fields and Build the CHAID Churn Model



Activity. A Type node defines how each field is used in modeling, including setting churn as the target field and marking appropriate customer attributes as inputs. The CHAID modeling node is then used to create a decision tree.

Purpose and expected outcome. The tree represents patterns associated with churn and can be used to estimate which customers may leave the service. The visible tree and predictor details are illustrative and should not be treated as verified model results.

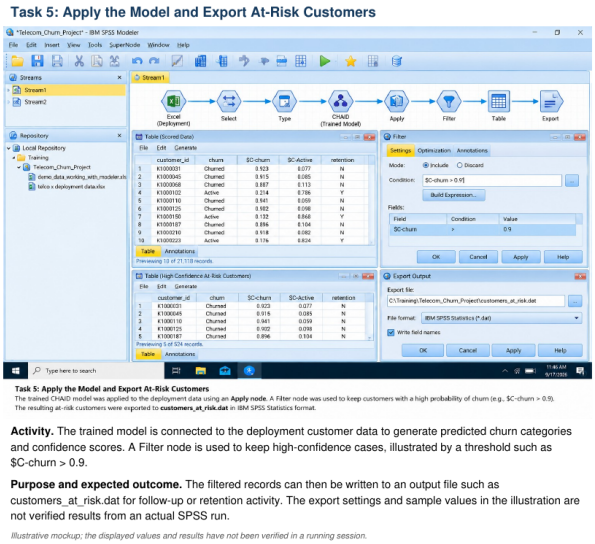
Illustrative mockup; the displayed values and results have not been verified in a running session.

Objective: Prepare the field roles and train a decision-tree model to identify patterns associated with customer churn.

Procedure: Add a Type node after the selected data. Review each field's measurement level and role, set `churn` as the target, and keep suitable customer attributes as input fields. Then add a CHAID modeling node, connect it to the prepared data, review its settings, and run model generation.

How it is used: CHAID builds a decision tree by splitting records into groups based on relationships between the target and predictor fields. Review the generated model and its predictor splits before using it. The model should be evaluated with appropriate data and metrics; a tree display alone does not establish that predictions are accurate.

Task 5: Score Customers and Export the At-Risk List



Objective: Apply the trained churn model to customer records and create a focused list of customers predicted to be at risk.

Procedure: Connect the generated model to the current or deployment customer data and run scoring to produce predicted churn values and confidence scores. Use a Filter node to retain high-confidence predictions; the tutorial describes a confidence threshold greater than 0.9. Confirm the exact generated field name in the stream before writing the filter expression, because model-output field names can vary by configuration.

Export: Connect an output node and save the filtered records to the specified output file, such as `customers at risk.dat`. Reopen or preview the exported file to verify that it contains the expected customer fields and predicted results. The resulting list can support retention follow-up, but predictions should be reviewed alongside business context before action is taken.